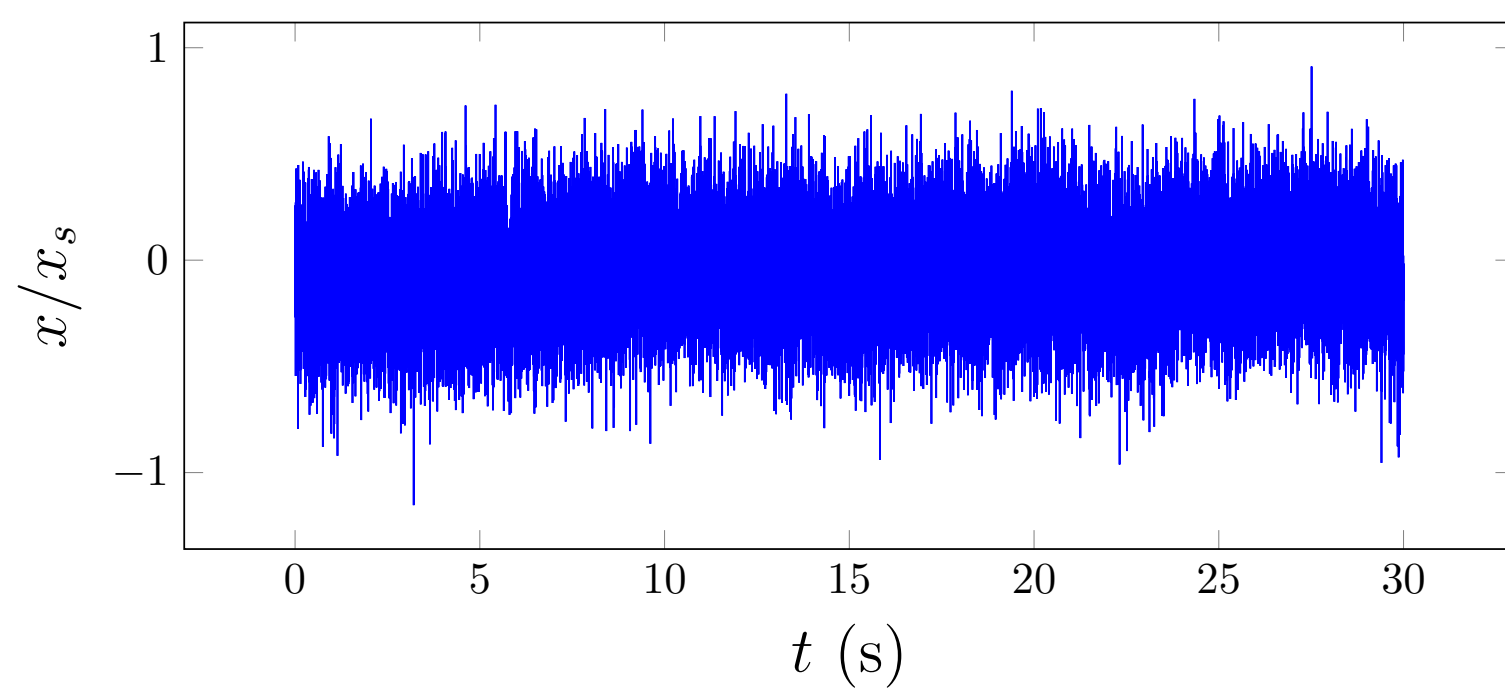


(a)

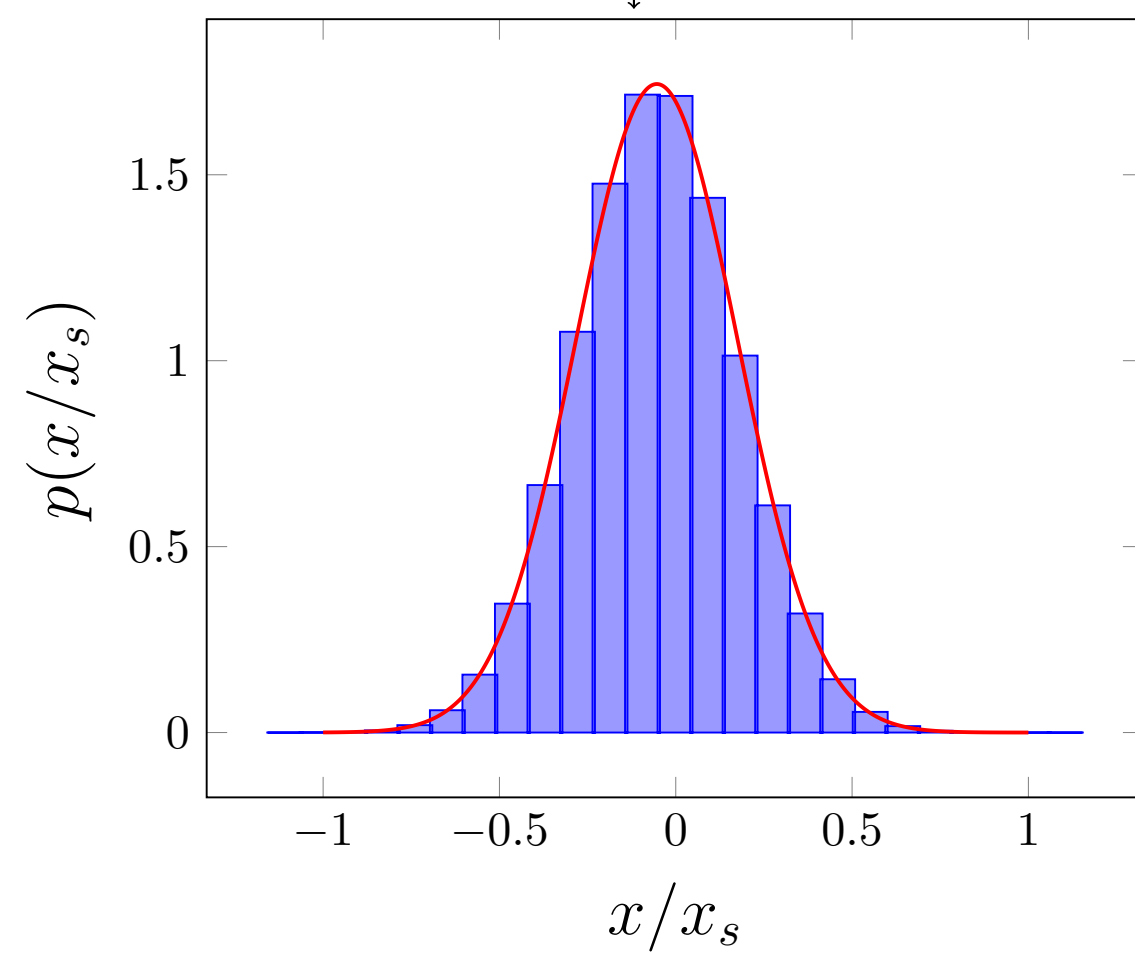


$$\frac{k_B T_c}{\sigma_x^2}$$

 $k_{\text{equipartition}}$

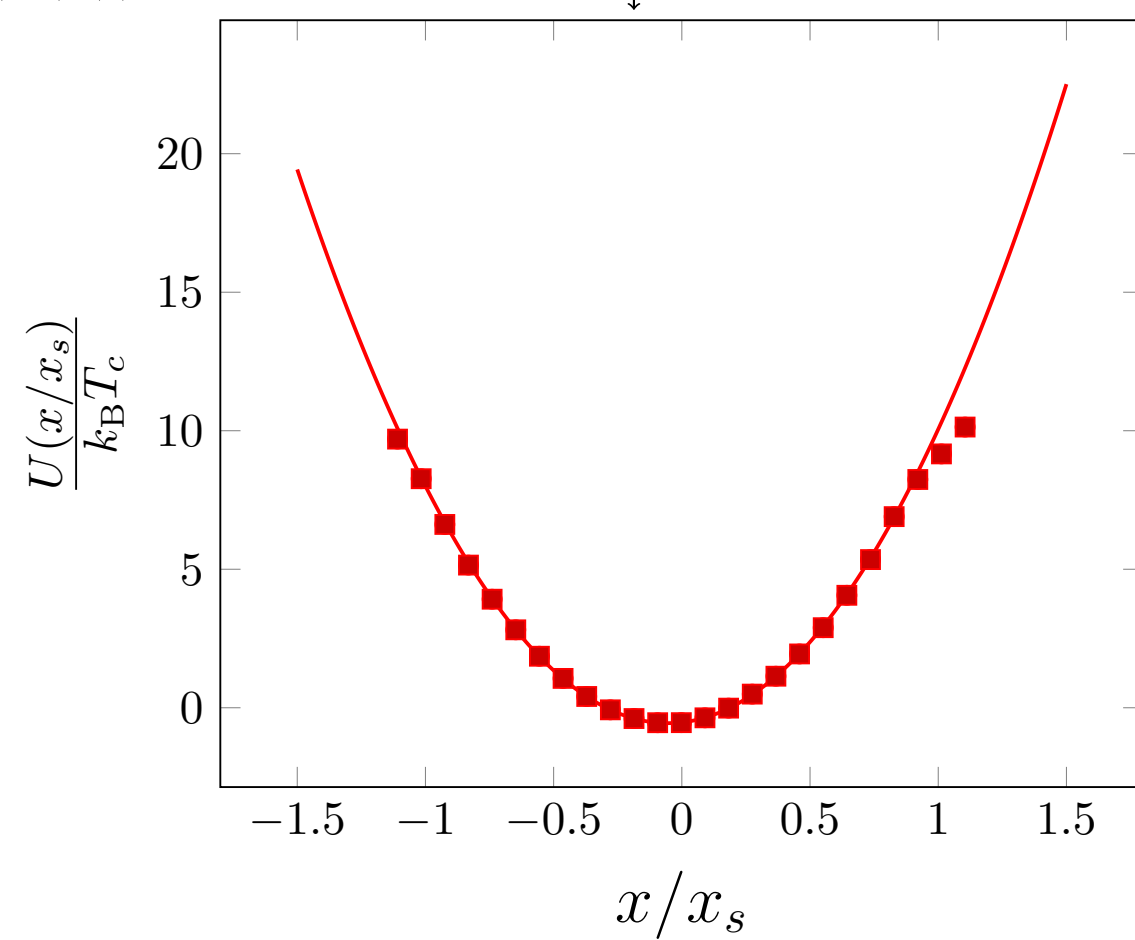
Histogram and normalise

(b(i))



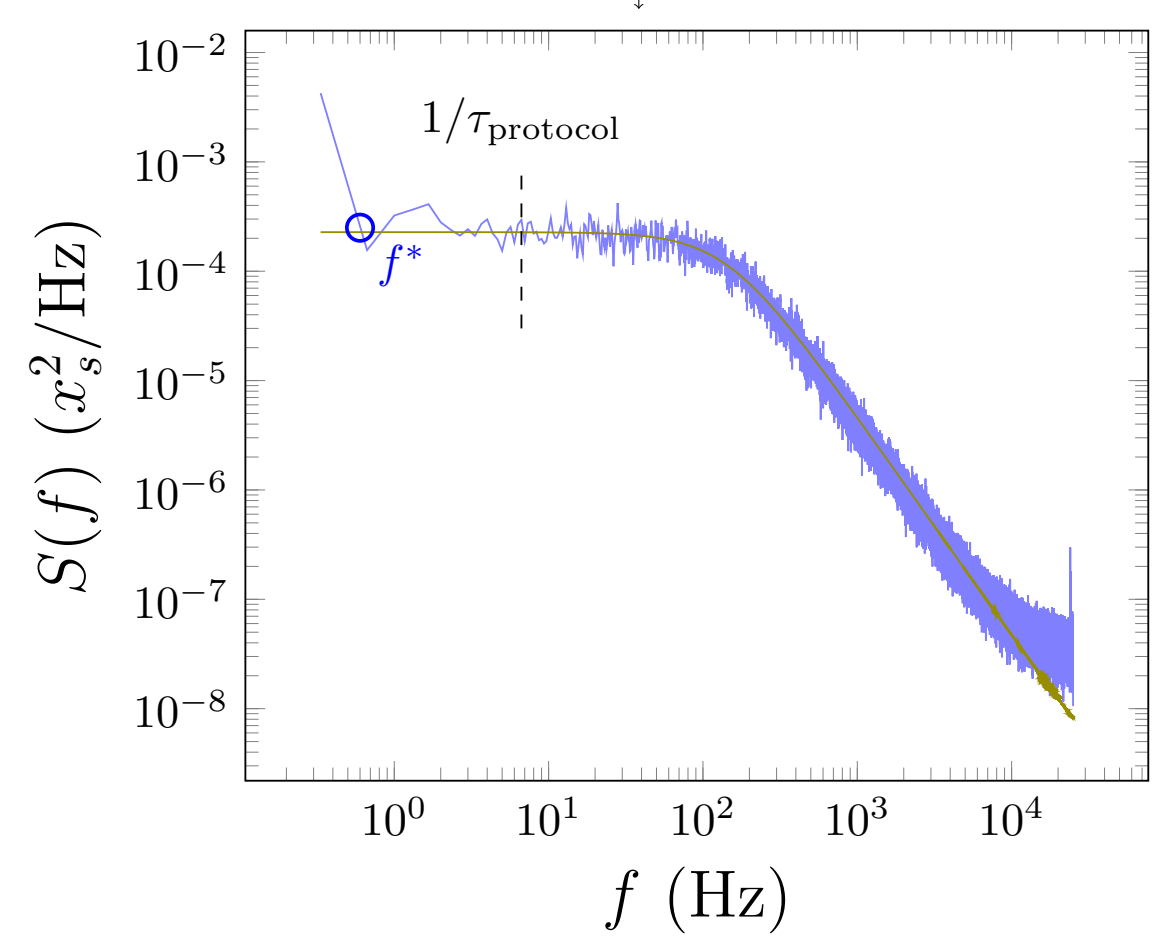
$$U(x) = k_B T_c \ln[p(x)]$$

(b(ii))


 k_{PDF}

$$S_n = \frac{1}{2\pi f_c} \frac{(1-c^2)D\Delta t}{1+c^2-2c\cos(2\pi n/N)}$$

(c)


 k_{PSD}